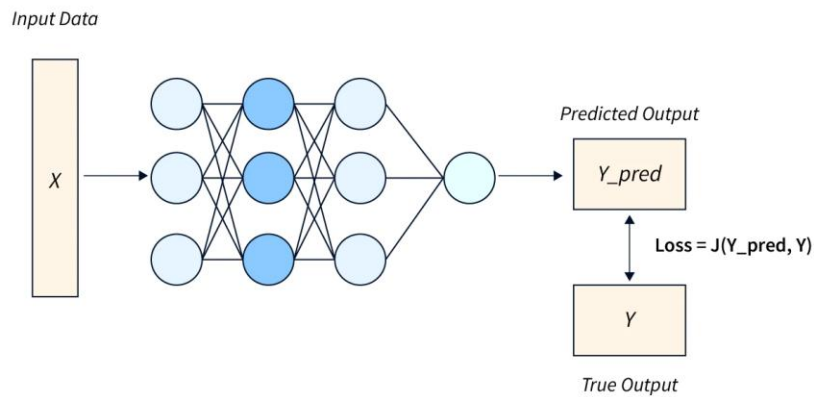


$$\text{MSE} = \frac{1}{N} \sum_i^N (Y_i - \hat{Y}_i)^2$$

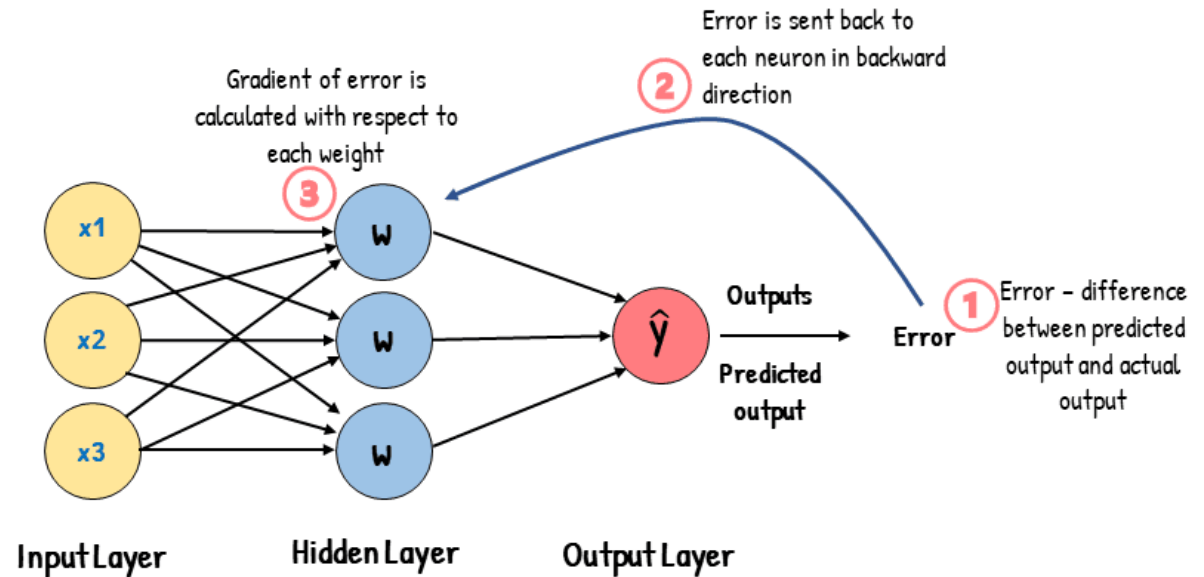
Loss Functions: Measuring the Error



- A loss function quantifies how "wrong" the model's prediction is compared to the actual target.
- **Mean Squared Error (MSE):** The standard for regression tasks (e.g., predicting house prices).
- **Categorical Cross-Entropy:** The standard for classification tasks (e.g., identifying digits); it measures the distance between probability distributions.

Backpropagation: Distributing the Blame

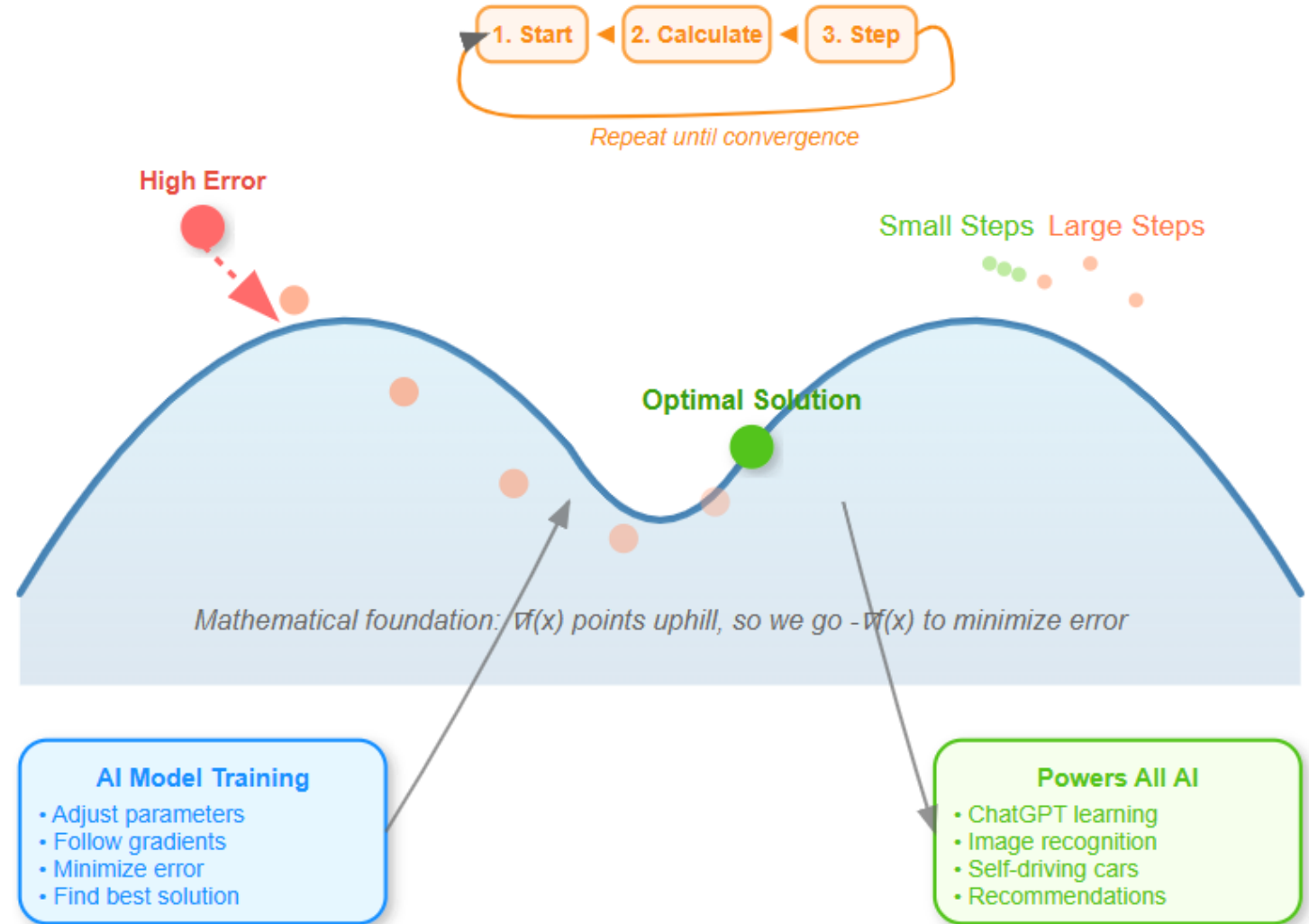
- Once we have a loss value, we need to know how to change the weights to reduce it.
- The **Chain Rule** from calculus is used to calculate the gradient of the loss with respect to every weight in the network.
- It effectively allows the error to flow backward from the output layer to the input layer.

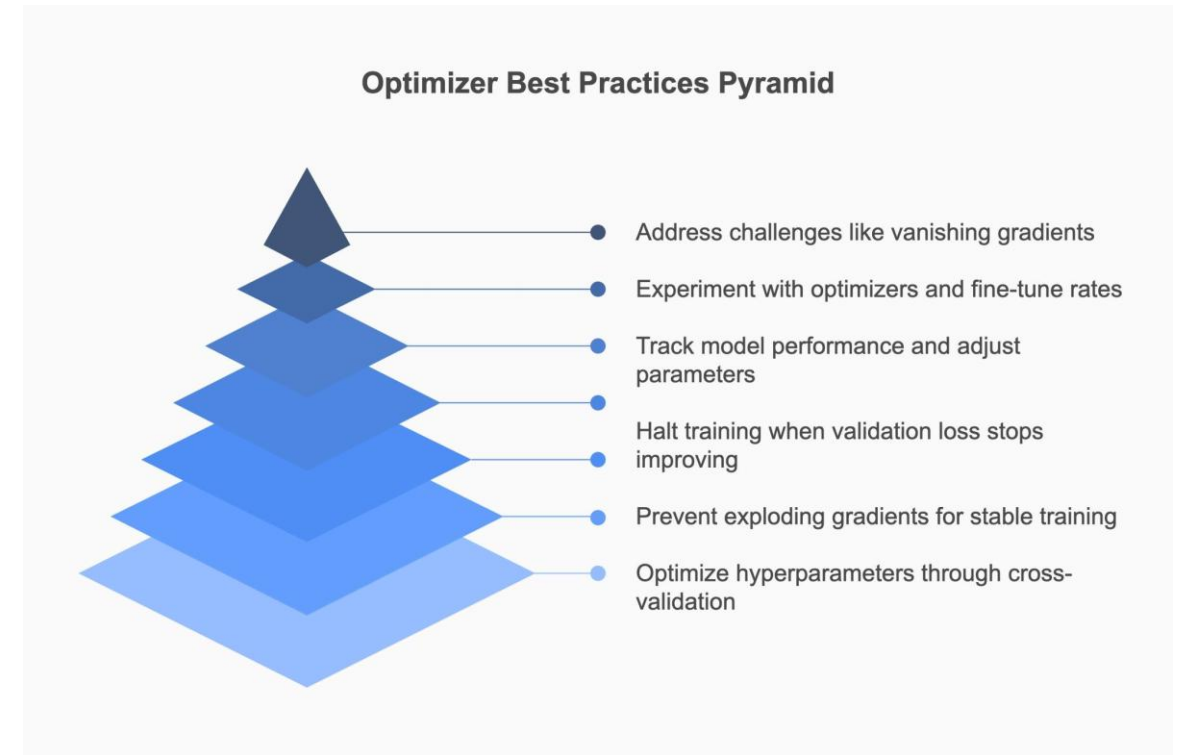
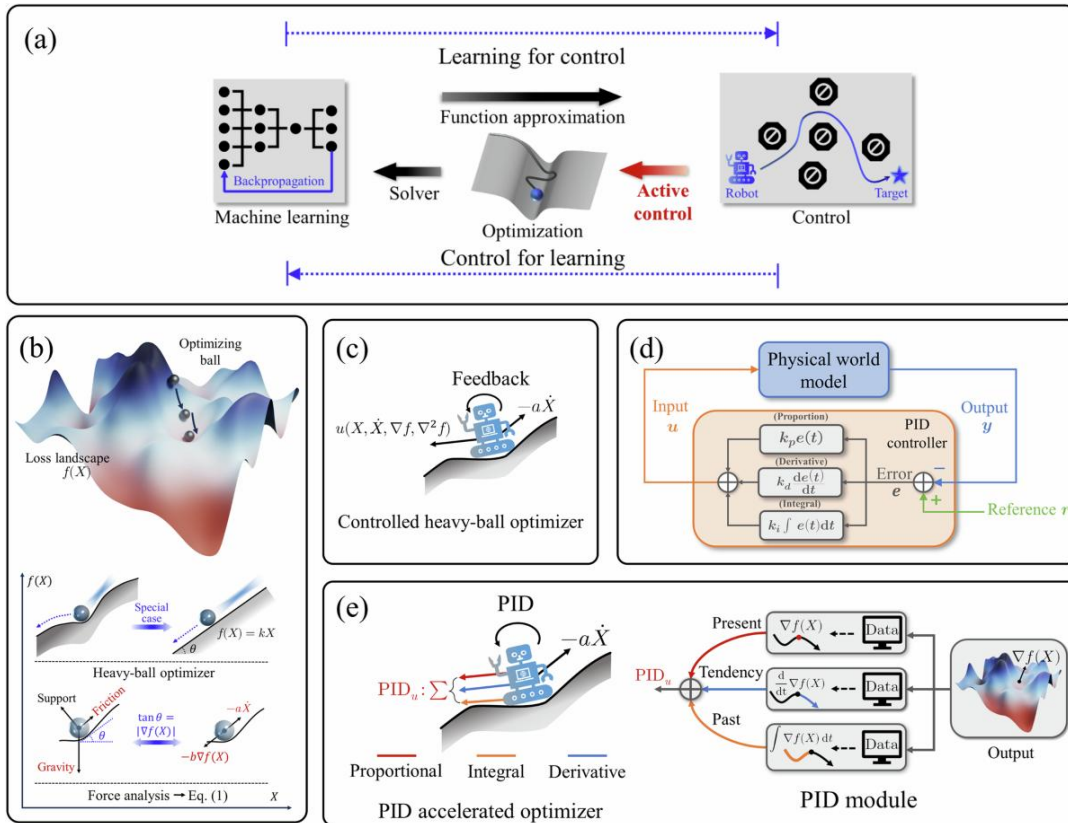


Gradient Descent: The Downhill Walk

- We update weights by taking a small step in the opposite direction of the gradient.
- **Stochastic Gradient Descent (SGD)**: Updates weights using one sample (or a small batch) at a time.
- **Learning Rate (η)**: A crucial hyperparameter that determines the size of the step; too big and you overshoot, too small and training takes forever.

How AI Systems Learn: Following Gradients Downhill





Advanced Optimizers: Speeding Up Convergence

- Basic SGD can be slow or get stuck in local minima.
- **RMSProp**: Adjusts the learning rate for each weight based on recent gradients.
- **Adam/AdamW**: The "gold standard"; it combines the benefits of momentum and adaptive learning rates for faster, more stable training.